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BSc (Hons) in Computer Science

Improving Mental Health Patient Prioritisation in NHS Services Using AI-Driven NLP Algorithms

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Abstract

NHS mental health services are under growing pressure from rising demand and limited resources, which increases the risk of delays in identifying patients at risk of self harm. Triage teams rely heavily on referral text, yet current processes to review this information are time consuming and can be inconsistent. This dissertation investigates how Natural Language Processing can support risk-based prioritisation of mental health referrals. An existing AI/NLP algorithm is enhanced and evaluated on anonymised referral style text, and then embedded in a prototype decision support tool that simulates an NHS triage workflow. The work aims to improve the detection of self-harm indicators, reduce manual workload, and inform the safe use of AI in mental health decision making.

Dedicated to those working in mental health services.

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Chapter 1

Introduction

1.1 Overview

Mental health services in the NHS face unprecedented demand, with increasing numbers of patients requiring urgent assessment and intervention. The timely and accurate identification of patients at high risk of self-harm is critical to patient safety and clinical outcomes. Yet current triage processes are often manual, time-consuming, and subject to inconsistency.

NHS mental health triage teams must process large volumes of referral letters and clinical notes in limited time, which increases the risk that subtle indicators of self-harm are overlooked. Advanced NLP algorithms can offer a consistent, data-driven approach to flagging patients at highest risk, thereby supporting more efficient and equitable triage. While NLP methods for risk detection have been validated in various healthcare contexts, few studies have specifically focused on enhancing existing algorithms and deploying prototypes within simulated NHS workflows. This dissertation addresses that gap by combining technical model enhancement, prototype development, and rigorous evaluation in a setting that reflects real-world constraints and clinical decision-making processes. By exploring how artificial intelligence and natural language processing can enhance the prioritisation of mental health patients through the automated detection of self-harm risk indicators within clinical referral text, this work contributes both methodologically and practically to the growing field of AI-augmented clinical decision support.

Beyond the technical focus, this research is motivated by the practical realities of frontline services. Any proposed system must be usable by clinicians who may not have specialist technical training, must integrate with existing NHS workflows, and must respect the sensitivities of working with vulnerable patients. The introduction therefore not only frames a computational challenge, but also sets out a broader question of how AI tools can be designed in a way that genuinely supports, rather than disrupts, everyday clinical practice.

1.2 Problem Statement

To determine patients' immediate need for specialist intervention, current manual review relies on clinician interpretation of free text, which is vulnerable to fatigue, time pressure, and subjective variation. As a result, high-risk patients may be missed or deprioritised, while lower-risk cases may consume disproportionate resources.

In addition, existing AI approaches in this area are often evaluated on curated research datasets that do not fully reflect the messiness of real-world referral text or the constraints of NHS settings. There is limited work that takes an existing algorithm, systematically enhances it, and embeds it into a realistic triage scenario where practical considerations such as response

time, interpretability, and clinician trust are taken into account. This creates a gap between promising research prototypes and tools that could plausibly be adopted in routine clinical workflows.

1.3 Research Questions

This research intends to answer the following questions:

1. What is the capability of Natural Language Processing (NLP) models to identify indicators of self-harm risk within mental health referral or clinical text?
2. How can enhancing existing AI algorithms improve the accuracy and reliability of patient prioritisation in NHS mental health triage systems?
3. Why is it ethically and practically significant to integrate AI-driven risk assessment tools into clinical decision-making for mental health services?

1.4 Research Objectives

1. To investigate current methods used for prioritising mental health patients in NHS triage.
2. To develop an AI-driven NLP algorithm to improve the detection of self-harm risk in referral text and patient prioritisation.
3. To evaluate the enhanced algorithm and decision-support prototype in a simulated NHS triage scenario.

1. *To review and analyse current methods used for prioritising mental health patients in NHS triage, so that the limitations and requirements for an improved NLP-based approach are clearly understood.*
2. *To design and develop an AI-driven NLP algorithm to improve the detection of self-harm risk in referral text and patient prioritisation, so that risk signals are identified more consistently than with current manual processes.*
3. *To implement and evaluate the enhanced algorithm and decision-support prototype in a simulated NHS triage scenario, so that its performance and practical feasibility for clinical use can be assessed.*

1.5 Scope

This research focuses specifically on benchmarking and enhancing NLP-driven risk assessment frameworks for patient prioritisation in NHS mental health triage, demonstrated through the automated analysis of referral text for self-harm indicators. The approach addresses the challenges of clinical decision support in a high-stakes, data-rich environment and evaluates the practical reliability of a prototype system within simulated NHS workflows.

Because direct integration and testing against live NHS systems is not feasible within the constraints of an undergraduate project, this study instead constructs a simulated triage environment. Referral-style text is collected and processed in a way that mimics NHS workflows, allowing the technical approach to be evaluated while respecting data access, ethical, and safety constraints.(new)

1.6 Ethical Considerations

Key ethical considerations include the following:

- Patient safety risks arising from false negative predictions that may fail to flag high-risk patients, or false positives that may unnecessarily escalate cases.
- Transparency and explainability of model outputs to ensure clinicians understand how risk assessments are generated.
- Bias and fairness in algorithm performance across diverse patient demographics and clinical presentation styles.
- Data privacy and confidentiality addressed through the use of anonymised or synthetic referral text, with strict adherence to NHS data governance and GDPR requirements.
- Automation bias mitigation by designing the tool as decision support rather than decision replacement.
- Informed consent and stakeholder engagement to ensure that patients, clinicians, and NHS staff understand how AI is used in triage processes.

1.7 Dissertation Outline

Chapter 2 presents a comprehensive literature review, covering existing NLP applications across diverse domains, current use of AI in NHS mental health, and technical benchmarks. It also identifies specific methodological and practical gaps in the literature that directly inform the design of this study. Chapter 3 outlines the research methodology, including data sources, preprocessing, model development, and evaluation protocols, with a focus on reproducibility and alignment with real NHS constraints. Chapter 4 presents results and discussion of the enhanced algorithm and prototype tool, linking quantitative performance metrics to their potential impact on triage decisions. Chapter 5 concludes with reflections on the broader implications for AI deployment in mental health services and recommendations for future research and implementation.

Chapter 2

Literature Review

2.1 Introduction

This chapter reviews the current state of artificial intelligence and natural language processing as applied to healthcare decision support, with particular focus on mental health triage and self-harm risk detection. The review examines technical benchmarks, existing NHS initiatives, ethical considerations, and gaps in the literature that motivate the present work. Give a overview of the sections in Lit Review for example Section 2.2 discuss NLP techhnology

2.2 Natural Language Processing and Large Language Models

This section introduces the key concepts and technologies in Natural Language Processing and large language models that underpin the models developed later in Chapters 3 and 4.

Artificial intelligence refers to computational systems that can perform tasks typically requiring human cognition, such as pattern recognition, decision making, and language understanding (Morandín-Ahuerma, 2022). Within AI, Natural Language Processing is the subfield concerned with enabling computers to interpret, analyse, and generate human language in a meaningful way. NLP combines techniques from linguistics, computer science, and statistics to extract structured information from unstructured text, such as clinical notes, referral letters, or patient communications.

Modern NLP relies heavily on machine learning, where algorithms learn patterns from large datasets rather than following hand-coded rules. Early NLP systems used simple keyword matching or rule-based grammars, but these struggled with the complexity and variability of real-world language (Manning, 2015). The introduction of statistical machine learning methods, such as Support Vector Machines and Naive Bayes classifiers, allowed systems to learn from annotated examples and generalise to new text. More recently, deep learning architectures, particularly neural networks with multiple layers of representation, have dramatically improved NLP performance across a wide range of tasks (Le Glaz et al., 2021).

The most significant recent advance has been the development of transformer-based models such as BERT (Bidirectional Encoder Representations from Transformers), which use self-attention mechanisms to capture contextual relationships between words in long passages of text (Devlin, Chang, Lee, and Toutanova, 2019). These pre-trained models can be fine-tuned on domain-specific data, such as clinical records, to perform tasks like entity recognition, sentiment analysis, and document classification with high accuracy (Zhang, Schoene, Ji, and Ananiadou, 2022). In healthcare, this capability is particularly valuable for processing the large volumes of

unstructured clinical notes generated daily, enabling automated extraction of diagnoses, symptoms, medications, and risk indicators that would otherwise require manual review (Ahmed, Kornblum, Oliver, Fusar-Poli, and Patel, 2023).

2.3 AI and NLP in Healthcare Decision Support

AI and NLP have become practical tools in healthcare settings where clinical decision making depends on large volumes of textual information. The ability to extract structured meaning from unstructured clinical notes, referral letters and patient records has moved from research into operational deployment, particularly in health systems facing rising demand and limited clinical resources.

The NHS has been at the forefront of investigating these methods. Reviews of AI-driven digital interventions in mental health now show that applications such as screening, triage and risk assessment are technically achievable and, in early pilots, deliver measurable gains in consistency and efficiency (Ni and Jia, 2025). These systems rely on machine learning models trained to recognise patterns in clinical language that correspond to specific concerns. When deployed properly, they can highlight cases requiring urgent attention and reduce the manual workload that triage teams face.

Several initiatives have tested this approach in NHS mental health services. The Chronosig project developed a BERT-based tool to prioritise referrals to secondary mental health services, demonstrating that automated analysis of referral text could support faster and more consistent decision making (Health Innovation Oxford, 2024). Multi-hospital studies using transformer-based models have also validated these techniques for detecting self-harm risk across different sites (Yang et al., 2023). The core method is similar in each case: pre-trained language models are fine-tuned on clinical text to identify indicators of risk such as expressions of suicidal ideation or mentions of prior self-harm.

The promise of NLP in this context is clear. Clinical triage relies heavily on subjective interpretation of referral text, which can vary between clinicians and is vulnerable to fatigue or time pressure. An automated system offers a consistent first pass that highlights linguistic signals of risk. However, the technology has limitations. Models trained on one trust's data may not generalise well to another, and free-text notes vary widely in format, completeness and language style (BMJ Open, 2023). There is also the risk of automation bias, where clinicians over-rely on algorithmic outputs and fail to apply their own judgement.

Despite these challenges, recent evidence suggests that AI and NLP can improve healthcare decision support when deployed with appropriate safeguards. The emphasis must remain on supporting rather than replacing clinical decision making, ensuring that models are explainable and that clinicians retain final responsibility for patient care.

2.3.1 Evolution of Clinical NLP Applications

Clinical Natural Language Processing has evolved from relatively simple rule-based systems into sophisticated deep learning and transformer-based models that can operate at scale across entire health records. Early work in clinical NLP largely relied on hand-crafted rules and dictionaries to identify key terms such as diagnoses, medications, and symptoms in free text. The next major step was the adoption of traditional machine learning approaches such as Support Vector Machines and Naive Bayes. Systematic reviews report that such models could reach reasonable performance, with many studies achieving accuracy or AUC values in the 0.70 to 0.85 range for mental health-related tasks (Le Glaz et al., 2021).

The arrival of deep learning brought substantial improvement. Convolutional and recurrent

neural networks (CNNs and LSTMs) were used to learn dense representations of text directly from data. In mental health and risk prediction, LSTM models were applied to sequences of notes and encounters in electronic health records to detect suicidal ideation or future self-harm attempts, typically reporting higher F1 scores and AUCs compared to traditional baselines (Le Glaz et al., 2021).

The emergence of transformer-based models such as BERT marked another turning point. Transformers use self-attention mechanisms to capture relationships between all tokens in a sequence simultaneously, which is particularly useful for long, complex clinical narratives. Fine-tuned transformers have been shown to outperform older approaches in high-stakes document review tasks across legal, financial, and healthcare domains (Pai et al., 2023; Li, 2023; Le Glaz et al., 2021).

In parallel, specialised clinical NLP toolkits have matured. Systems such as MedCAT provide concept recognition and linking to clinical ontologies, enabling large-scale extraction of symptoms, diagnoses, and risk factors from routine clinical notes. Applied within the NHS, such pipelines have been used to analyse electronic health records for remote mental health care and crisis prediction (BMJ Open, 2023).

Most recently, there has been growing interest in adapting large general-purpose language models to clinical tasks. However, their use in mental health decision support remains cautious, due to concerns about hallucination, transparency, and data governance. Many current NHS-oriented projects focus on fine-tuned, task-specific transformers combined with robust evaluation, rather than direct deployment of general-purpose LLMs.

Case Study: NLP-Based Triage in NHS Mental Health Services

Recent NHS-oriented research has moved beyond isolated model evaluation toward end-to-end triage pipelines that integrate NLP models into referral workflows. One prominent example is the Chronosig project, which developed a transformer-based system to prioritise referrals to secondary mental health services using free-text clinical information. The pipeline combines automated language analysis with clinician oversight, reflecting the practical constraints and ethical requirements of NHS triage.

As shown in Figure 2.1, referral text and historical clinical information are processed using a fine-tuned transformer model to generate a prioritisation score, which is then reviewed by clinicians rather than acted upon autonomously.

2.3.2 Existing Applications and Benchmarks

AI-driven NLP has found extensive application across industries, often demonstrating capabilities that parallel or exceed those in mental health. Table 2.1... In customer service, transformer-based architectures such as BERT capture complex linguistic patterns to classify customer feedback with high accuracy (Ismail et al., 2024). The legal profession uses Technology-Assisted Review (TAR) to sift through documents, with research confirming that TAR is both more efficient and effective than exhaustive manual review (Grossman and Cormack, 2011). Recent studies have fine-tuned open-source LLMs to improve performance in predictive coding tasks (Pai et al., 2023). Financial fraud detection applies NLP to analyse unstructured text from financial reports, with BERT models achieving over 83% accuracy (Li, 2023). These systems often use a human-in-the-loop workflow, which parallels the proposed model for clinical triage.

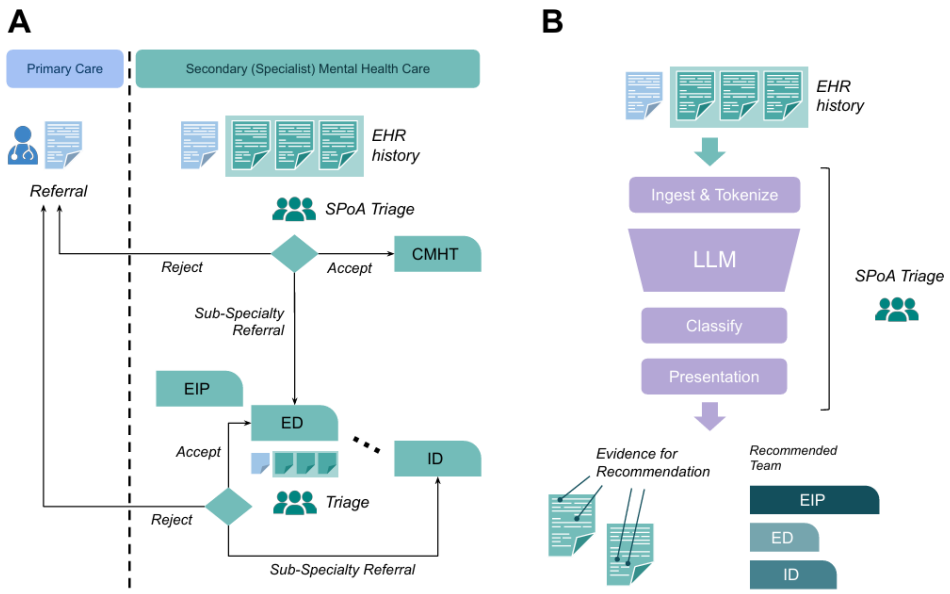

Figure 2.1: End-to-end NLP-driven referral triage pipeline used in NHS mental health services, adapted from (Taylor, Kormilitzin, Lorge, Nevado-Holgado, and Joyce, 2024) and the Chronosig project.

Source	Application Domain	Algorithms / Methods	Evaluation Metrics	Performance Scores
Ismail et al. (2024)	Customer Service	BERT, LSTM (Sentiment Analysis)	Accuracy, Precision, Recall, F1	**85–95%** accuracy.
Grossman and Cormack (2011)	Legal (e-Discovery)	Technology-Assisted Review (TAR) using SVMs	Recall, Precision, Elusion	**>75%** recall while reviewing **<5%** of documents.
Pai et al. (2023)	Legal (e-Discovery)	Open-source LLMs (e.g., FLAN-T5)	Precision, Recall, F1	**F1-Score of 0.79** with minimal training examples.
Li (2023)	Financial Fraud Detection	CNN, LSTM, BERT applied to 10-K reports	Accuracy, Precision, Recall, F1	BERT achieved **83.33%** accuracy.
Le Glaz et al. (2021)	Mental Healthcare	Systematic review of SVM, CNN, LSTM, BERT	Accuracy, AUC, Precision, Recall	Model performance ranges from **0.70 to >0.95**.

Table 2.1: Comparative Analysis of NLP Applications Across Example Domains for slected reserachs

AI-Supported Referral Prioritisation in Clinical Practice

Beyond mental health, similar NLP-driven prioritisation pipelines have been implemented in other areas of clinical triage. (Abdel-Hafez et al., 2023) propose an AI-assisted referral workflow based on Clinical Prioritisation Criteria (CPC), where machine learning models classify referral urgency from unstructured text in order to support scheduling and resource allocation. Although applied outside mental health, the architecture closely mirrors NHS triage processes.

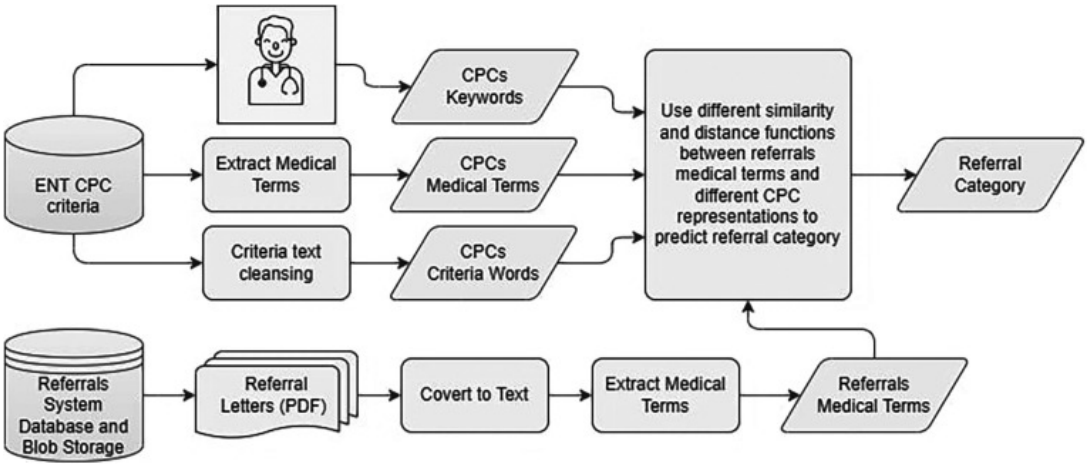

Figure 2.2: AI-assisted referral prioritisation workflow using Clinical Prioritisation Criteria (CPC), adapted from (Abdel-Hafez et al., 2023).

Figure 2.2 illustrates how machine learning models are embedded within an established clinical workflow rather than operating as standalone predictors. This human-in-the-loop design aligns with NHS governance requirements and highlights common architectural principles shared with mental health triage systems, despite differences in clinical domain.

2.3.3 Limitations of Existing NLP-Based Triage Systems

Despite demonstrating the feasibility of NLP-driven mental health triage, existing end-to-end systems such as ChronoSig and CPC exhibit several documented limitations related to performance robustness, accuracy, and clinical applicability.

Reported evaluations of Chronosig indicate strong aggregate performance on retrospective referral datasets; however, external validation across different NHS trusts and diagnostic presentations is ongoing. Performance is expected to vary across risk categories and clinical presentations, particularly for patients with less explicit or indirectly expressed indicators of distress, as is common in atypical presentations and certain demographic groups (Joyce et al., 2021). This suggests a reliance on overt linguistic markers, limiting effectiveness in cases where risk is conveyed implicitly or through temporal context.

Similarly, CPC-based architectures achieve competitive classification metrics but are constrained by annotation uncertainty. Risk labels are derived from clinician judgement rather than objective outcomes, introducing label noise that directly affects achievable accuracy (Sylolypavan, Sleeman, Wu, and Sim, 2023). Inter-rater variability in mental health assessment further complicates performance interpretation, as misclassifications may reflect ambiguity in ground truth rather than model error.

Both systems prioritise recall to minimise false negatives, a design choice aligned with patient safety objectives. However, published results indicate elevated false positive rates, which may reduce practical utility by increasing clinician workload and diminishing trust in automated triage recommendations (Brann, Sterling, Frisch, and Schrager, 2024). Such trade-offs highlight limitations in precision that are not fully captured by aggregate F1 or AUC metrics.

Generalisation remains a further challenge. Existing benchmarks are typically trained and evaluated on datasets sourced from a limited number of clinical contexts. Prior studies caution that transformer-based models may overfit stylistic or institutional artefacts, resulting in degraded accuracy when applied to unseen healthcare settings or populations (Cohen et al., 2022).

Finally, current evaluations rely predominantly on offline performance metrics such as precision, recall, and AUC. While suitable for algorithmic benchmarking, these measures do not capture downstream clinical impact, user interaction effects, or decision support effectiveness in real-world settings. Prior studies in clinical AI caution that strong offline performance does not necessarily translate into improved decision-making or workflow integration, leaving open questions regarding operational reliability and practical accuracy (Kelly, Karthikesalingam, Suleyman, Corrado, and King, 2019; Topol, 2019).

Collectively, these limitations indicate that while existing NLP-based triage systems establish a strong foundation, their reported performance and accuracy remain constrained by data quality, annotation subjectivity, and limited generalisability. These gaps motivate further investigation into improved risk feature extraction and calibration for NHS mental health settings.

2.4 Self-Harm Risk Detection in Mental Health

The detection of self-harm risk from clinical text is a critical application of NLP. Studies using transformer-based and deep learning models have shown that patterns of language in referral letters and clinical notes can reliably predict patients at elevated risk. These approaches typically combine entity recognition (identifying mentions of prior self-harm, suicidal ideation, psychiatric medications, etc.) with document-level classification to assign a risk score.

The challenge lies in balancing sensitivity (catching all true high-risk cases) with specificity (minimising false alarms). In clinical contexts, false negatives (missing a high-risk patient) carry severe consequences, while false positives consume triage resources. Model development must therefore prioritise recall, often at the expense of precision, and results must be presented to clinicians with uncertainty estimates and supporting evidence (i.e., which linguistic features contributed to the risk score).

2.5 Technical Benchmarks and Evaluation

Modern NLP model evaluation relies on standard train/validation/test splits (typically 80/10/10) and k-fold cross-validation. The primary metrics are Precision, Recall, F1-Score, and AUC-ROC. In healthcare applications, sensitivity and specificity are also important, as they directly relate to clinical utility and the balance between missed high-risk cases and unnecessary alarms (Le Glaz et al., 2021).

Pre-trained transformer architectures such as BERT and RoBERTa are commonly fine-tuned on domain-specific text data using deep learning frameworks like PyTorch together with the Hugging Face Transformers library (Wolf et al., 2020; Hugging Face, 2024a). These models are computationally intensive and require careful hyperparameter tuning (learning rate, batch size, number of epochs) and early stopping to prevent overfitting. In mental health research, fine-tuned BERT models have been used to classify online posts or clinical text into suicidal versus non-suicidal categories, achieving strong performance on standard metrics (Ji et al., 2018).

2.5.1 Implementation with PyTorch and Transformers

The purpose of this subsection is not to propose a new algorithm, but to illustrate a typical fine-tuning pipeline for BERT-style models, so that later methodological choices in Chapter 3 are clear even to readers without day-to-day experience in deep learning.

In practice, fine-tuning a pre-trained BERT model for mental health risk classification follows a standard workflow demonstrated in the official Transformers examples and PyTorch tutorials

(Hugging Face, 2024b; PyTorch, 2024). The process begins by loading a pre-trained model and its associated tokeniser:

```

from transformers import BertTokenizer,
    BertForSequenceClassification

model_name = 'bert-base-uncased'
tokenizer = BertTokenizer.from_pretrained(model_name)
model = BertForSequenceClassification.from_pretrained(
    model_name,
    num_labels=2  # high risk vs not high risk
)
```

Listing 2.1: Loading a Pre-trained BERT Model

In listing 2.1 the model is not trained from scratch. Instead, a general-purpose BERT model is loaded and adapted for a simple two-class problem (high risk versus not high risk). Fine-tuning then adjusts the weights of this model using referral text so that its decisions reflect the mental health triage task.

Clinical referral text is then tokenised into input IDs and attention masks suitable for transformer models:

```

text = "Patient reports persistent suicidal ideation and recent self
    -harm."

inputs = tokenizer(
    text,
    padding=True,
    truncation=True,
    max_length=512,
    return_tensors='pt'
)

# inputs['input_ids'] and inputs['attention_mask'] are passed to the
    model
```

Listing 2.2: Tokenising Referral Text

Listing 2.2 showcases the raw sentence is converted into numerical tokens and an attention mask so that it can be processed by the transformer model. In practice, the same procedure is applied to every referral in the dataset, producing consistent input representations for training and evaluation.

Training uses minibatches of tokenised examples and an optimiser such as AdamW, following the pattern in the Hugging Face text classification examples (Hugging Face, 2024b):

```
from torch.utils.data import DataLoader
from transformers import AdamW

train_loader = DataLoader(train_dataset, batch_size=16, shuffle=True
    )
optimizer = AdamW(model.parameters(), lr=2e-5)

model.train()
```

```python
for epoch in range(3):
    for batch in train_loader:
        input_ids = batch['input_ids']
        attention_mask = batch['attention_mask']
        labels = batch['labels']

        outputs = model(
            input_ids=input_ids,
            attention_mask=attention_mask,
            labels=labels
        )
        loss = outputs.loss

        optimizer.zero_grad()
        loss.backward()
        optimizer.step()
```

Listing 2.3: Simplified Fine-tuning Loop

Listing 2.3's loop repeatedly shows batches of referral examples to the model, compares its predictions with the true labels, and nudges the model parameters to reduce the loss. Hyperparameters such as batch size, learning rate, and number of epochs control how quickly the model learns and how likely it is to overfit the training data.

After training, the model is evaluated on a held-out test set. Predictions are converted to labels and standard metrics are computed using libraries such as scikit-learn, as is common in published work on suicidal ideation detection (Ji et al., 2018):

```python


from sklearn.metrics import classification_report, roc_auc_score

model.eval()
predictions, true_labels = [], []

with torch.no_grad():
    for batch in test_loader:
        input_ids = batch['input_ids']
        attention_mask = batch['attention_mask']
        labels = batch['labels']

        outputs = model(input_ids=input_ids,
                        attention_mask=attention_mask)
        logits = outputs.logits
        preds = logits.argmax(dim=1)

        predictions.extend(preds.cpu().numpy())
        true_labels.extend(labels.cpu().numpy())

print(classification_report(true_labels, predictions,
      target_names=['Not High Risk', 'High Risk']))
auc = roc_auc_score(true_labels, predictions)
print(f"AUC-ROC: {auc:.3f}")
```

Listing 2.4: Evaluation and Metric Calculation

Once training is complete, the model is tested on data it has not seen before. The code computes standard classification metrics such as precision, recall, F1-score, and AUC-ROC, which provide an objective summary of how well the model distinguishes high-risk from lower-risk referrals.

This workflow closely mirrors the implementations described in mental health NLP studies that fine-tune BERT for risk classification tasks, where reported AUC and F1-Score values typically exceed those of traditional machine learning baselines (Ji et al., 2018).

2.6 Integration into NHS Workflows and Ethical Governance

While the previous sections focused on model architectures and performance metrics, successful integration of AI into NHS triage also depends on organisational, ethical, and governance factors. Technical accuracy alone is not sufficient; stakeholder buy-in, robust data governance, and mitigation of known risks are equally important. Clinicians must understand how the algorithm generates scores, and the system must be designed as decision support (highlighting candidates for urgent review) rather than decision replacement (automatically determining triage priority).

Data governance is paramount. All patient data used for training, validation, and testing must be anonymised, stored securely, and accessed only by authorised personnel. Models must be regularly audited for bias, particularly across patient demographics, and performance must be continuously monitored in live settings to detect model drift or unintended failures.

Ethical considerations include the responsibility to avoid reinforcing historical biases in data, ensuring that false positives and false negatives are clearly communicated to clinicians, and maintaining transparency regarding model limitations.
These considerations build on the ethical themes outlined in Chapter 1 and frame the constraints within which any NLP-based triage prototype must operate in NHS mental health services.

Chapter 3

Methodology

This chapter describes the methodological approach adopted for the design, enhancement, and evaluation of an NLP-driven mental health triage system. This dissertation employs a mixed-methods approach, combining technical model development (quantitative) with qualitative evaluation of practical integration into simulated NHS workflows. The aim is to demonstrate that enhanced NLP algorithms can improve risk detection while remaining operationally feasible and aligned with real-world triage constraints.

3.1 System Design

The system proposed in this dissertation built on the end-to-end architectures described in aforementioned Figures 2.1 and 2.2. While these earlier pipelines demonstrated the feasibility of NLP-driven triage, they left open questions around enhanced risk feature extraction, model calibration, and explicit prioritisation logic in NHS mental health settings, which this work addressed.

Rather than connecting directly to live NHS referral systems, the prototype was implemented within a simulated triage environment. A simple web-based frontend was used to collect referral-like free-text descriptions and basic metadata, which were then processed by an AI-driven back-end model trained on anonymised or synthetic referral datasets. The model produced a calibrated risk score and suggested priority band that could be used to order cases within a clinician-facing triage queue.

3.1.1 High-level system architecture

Figure 3.1 illustrates the overall architecture of the AI-driven NLP prioritisation prototype. Incoming referral-style text and simple metadata were captured through a lightweight web or chat interface, forwarded to a FastAPI-based back-end, and passed to an NLP risk scoring module built around a BERT-based classifier. The resulting risk probability was then fed into a prioritisation and triage component that mapped scores to priority bands, generated a sortable prioritisation score, and exposed a prioritised referral list through a clinician triage interface for review rather than automatic action.

Figure 3.1: High-level architecture of the AI-driven NLP prioritisation prototype used in this dissertation.

3.1.2 Data pipeline for model training and evaluation

The full data pipeline for training and evaluating the prioritisation models is shown in Figure 3.2. Multiple referral-style free-text sources were combined, including anonymised datasets and open mental health corpora, together with synthetic triage labels and derived priority bands. These inputs were cleaned, de-identified, tokenised, and label-harmonised before being split into train, validation, and test sets. The splits were then used to train a baseline TF–IDF plus logistic regression model, an intermediate deep learning model, and an enhanced BERT-based classifier. On top of these models, a prioritisation logic layer performed probability calibration, threshold tuning, priority band mapping, and queue ordering, which were evaluated using both standard classification metrics and prioritisation-focused metrics such as the proportion of high-risk cases appearing near the top of the queue and the average rank of high-risk referrals.

Figure 3.2: Data pipeline for training, calibrating, and evaluating AI-driven prioritisation models for simulated NHS mental health triage.

3.1.3 Model architectures and prioritisation logic

Figure 3.3 compares the baseline and enhanced AI pipelines used in this study. The baseline route applied standard preprocessing and TF–IDF feature extraction followed by a logistic regression classifier that output a binary risk label, which was then mapped to a simple priority category and used to construct a sorted queue. In contrast, the enhanced route employed a BERT encoder and classification head to estimate a continuous risk probability, which was calibrated, thresholded into high, medium, and low priority bands, and then passed into a prioritisation module that produced an ordered triage list.

Figure 3.3: Comparison between the baseline TF–IDF plus logistic regression prioritisation pipeline and the enhanced BERT-based AI-driven prioritisation pipeline.

The internal structure of the enhanced model and prioritisation pipeline is expanded in Figure 3.4. Referral text and optional metadata were transformed into token IDs and attention masks by the BERT tokenizer, encoded into contextual embeddings by the BERT encoder, and passed through dropout and a linear classification layer to obtain logits. After applying softmax to obtain class probabilities, a calibration step on the validation set aligned predicted probabilities with observed risk, and tuned thresholds were used to assign priority bands. Finally, a prioritisation logic block converted the band and calibrated probability into a sortable score, applied tie-breaking rules where required, and output a risk probability, priority band, and prioritisation score for each referral.

Figure 3.4: Detailed BERT-based risk prediction and prioritisation pipeline, including calibration, thresholding, and queue scoring steps.

3.1.4 Simulated NHS triage workflow

To evaluate the system in a way that was compatible with the constraints of undergraduate research and NHS data governance, the prototype operated within a simulated mental health triage workflow, shown in Figure 3.5. On the patient or system side, referral-like text was

submitted, processed by the NLP risk model, and converted into a suggested priority band, after which the prioritisation module updated the triage queue and presented a prioritised list of open referrals. On the clinician side, staff accessed this list through a simple triage interface, inspected individual referrals together with the model-generated risk score, band, and brief explanation, then confirmed, upgraded, or downgraded the AI suggestion. The final triage decision was recorded as feedback for later analysis of model performance and misprioritised cases.

Figure 3.5: Simulated NHS mental health triage workflow showing how AI-driven NLP outputs were used to support, rather than replace, clinician prioritisation decisions.

prior art → gap → proposed system flow. (this high-level diagram flow of my system to be added later on as a final model adapted and considered to be developed.)

3.2 Data Collection and Preparation

The dataset will consist of anonymised or synthetic mental health referral text, sourced from NHS records or constructed to reflect realistic referral patterns. Text will be preprocessed (lowercased, tokenised, stripped of protected health information) and annotated with binary risk labels (high risk / not high risk) based on clinical consensus.

https://www.kaggle.com/code/vedikagupta0/mental-health-text-classification-analysis https://psresnik.github.io/umd_reddit_suicidality_dataset.html

3.3 Model Development

The baseline model will be an existing NLP classifier (e.g., logistic regression with TF-IDF features). Enhancements will include:

- Fine-tuning a pre-trained BERT or RoBERTa model on the referral text dataset.
- Integrating clinical entity recognition (using MedCAT or similar tools) to extract structured features.
- Ensemble methods combining multiple model architectures.

3.4 Evaluation and Prototype Testing

Models will be evaluated using standard metrics (Accuracy, Precision, Recall, F1-Score, AUC-ROC). The best-performing model will be embedded in a prototype decision-support tool and tested in simulated triage scenarios to assess usability and clinical relevance.

3.5 Limitations and Assumptions

Key limitations include limited access to real NHS data (necessitating synthetic data or case studies), small sample sizes, and inability to conduct prospective validation with live clinical users.

- The prototype operates in a simulated NHS triage environment using referral-style or chat-derived text rather than live NHS referral feeds. This limits the ecological validity

of the results and introduces potential differences in language style compared with real clinical documentation.

Chapter 4

Results and Discussion

4.1 Model Performance

[Results section to be completed with experimental findings, comparative benchmarks, and error analysis.]

4.2 Comparison with Literature

[Analysis of how results compare to state-of-the-art and existing NHS approaches.]

4.3 Limitations and Practical Implications

[Discussion of technical limitations, generalisability challenges, and real-world deployment considerations.]

Chapter 5

Conclusion

5.1 Summary of Contributions

5.2 Implications for NHS and Future Research

5.3 Final Remarks

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Appendices

Appendix A

Supplementary Materials

[Include additional tables, figures, code snippets, or ethics documentation as needed.]